

Basic Principles of Medicine 2

Module: Endocrinology and Reproduction.

DLA Video

Nutrition and Aging

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References

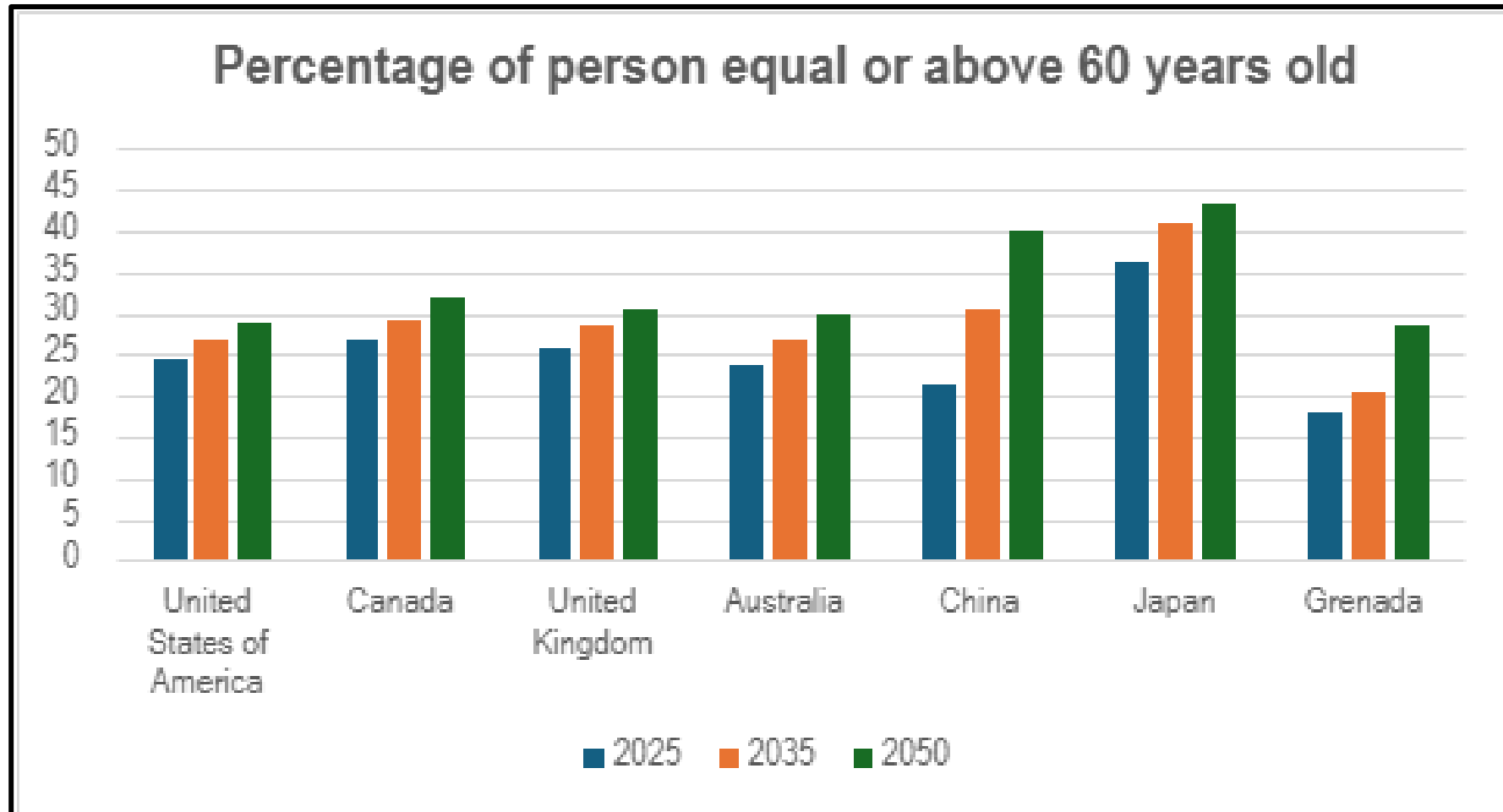
- 1) [Frailty - UpToDate](#)
- 2) [Geriatric nutrition: Nutritional issues in older adults - UpToDate](#)
- 3) <https://doi.org/10.3390/biomedicines8070198>
- 4) <https://platform.who.int/data/maternal-newborn-child-adolescent-ageing/ageing-data/ageing---demographics>
- 5) [Journal of Endocrinology](#) <https://doi.org/10.1530/JOE-16-0603>

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Objectives: DLA: Nutrition & Aging

SOM.MK.III.BPM2.2.ER.3.BCHM.1001	Describe changes in physiological, biochemical and endocrine functions associated with aging
SOM.MK.III.BPM2.2.ER.3.BCHM.1002	Identify the nutritional changes related to aging.
SOM.MK.III.BPM2.2.ER.3.BCHM.1003	Review nutrient requirements for the elderly including energy, protein, fat and cholesterol, carbohydrates, dietary fiber, water, vitamin B12, folate, calcium and vitamin D
SOM.MK.III.BPM2.2.ER.3.BCHM.1004	Describe the risk factors for poor nutrition in the elderly including poor appetite, immobility, concurrent illness, and decreased nutrient absorption
SOM.MK.III.BPM2.2.ER.3.BCHM.1005	Explain how under-nutrition exacerbates disease and leads to poor recovery from illness and hospitalization and describe how disease and hospitalization can lead to under-nutrition
SOM.MK.III.BPM2.2.ER.3.BCHM.1006	Review how the aging process can be accelerated by oxidative and inflammatory stress
SOM.MK.III.BPM2.2.ER.3.BCHM.1007	Define frailty and sarcopenia. Review metabolic changes associated with aging highlighting negative nitrogen balance and loss of muscle protein
SOM.MK.III.BPM2.2.ER.3.BCHM.1008	Review protein energy malnutrition in the elderly. Describe methods for identification and management

Population 60 years and above as a percentage of total population



Clinical case: Malnutrition in Elderly

- **Patient:** Mrs. Brittany Bones
- **Age:** 76
- **Sex:** Female
- **Occupation:** Retired schoolteacher
- **Marital status:** Widow
- **Living situation:** Lives alone in a single-level home with limited family support



Clinical case: Malnutrition in Elderly

Chief complaint: "My legs feel weak, and I can't eat like I used to"

History of Present Illness:

Mrs. James presents with progressive muscle weakness, particularly in the lower limb, over the past 2 years. She reports difficulty standing from seated position, climbing stairs and carrying groceries. She also describes a marked decline in her appetite over the last 6 months, with early satiety and weight loss of approximately 8kg. She occasionally feels dizzy when getting up and has had two minor falls in the last 3 months.

Past Medical History:

Hypertension, Osteoarthritis, Mild cognitive impairment.

Medications:

Amlodipine 5mg OD, Acetaminophen 500MG prn joint pain, Vitamin D3 1000 IU OD, Calcium carbonate supplement

Clinical case: Malnutrition in Elderly

Physical Examination:

General: Frail-appearing elderly woman, alert, oriented to person and place.

Height: 158 cm (5'2")

Weight: 48 kg (previously 56 kg)

BMI: 19.2 kg/m²

Muscle strength: 3/5 in proximal lower limbs

Gait: Slow, cautious, slight shuffling

Grip strength: Decreased



Endocrine and Physiological changes in aging

Hormonal changes	Effects on the elderly
Decreased growth hormone and IGF-1	Associated with lower strength and decreased mobility
Decreased dehydroepiandrosterone sulfate (adrenal androgen)	DHEA-S likely plays a direct role in maintaining muscle mass and indirectly prevents inflammatory pathways that contribute to muscle decline
Increased cortisol levels	May impact skeletal muscle and immune system components
Decreased sex steroids	Mixed evidence of reproductive hormones contribute to frailty
Decreased 25(OH) vitamin D	Decreased bone density however more pronounced in elderly females (low estrogen)
Decreased thyroid hormones	Decreased Basal metabolic rate

Biochemical and physiological changes in aging

Biochemical change	Effects in elderly
Decreased immune response	Increased serum levels of proinflammatory cytokine IL6 and CRP, as well as elevated white blood cell and monocyte counts.
Altered glucose metabolism	Increased risk of diabetes mellitus and cardiovascular disease
Dysregulation of the autonomic nervous system	Decreases digestion and nutrient absorption; Constipation
Chronic disease progression	Increased morbidity and mortality
Cognitive decline	Impacts behavior and social relationships

Changes in bone, muscle and fat tissues with increasing age and accompanying increase in low-grade inflammation

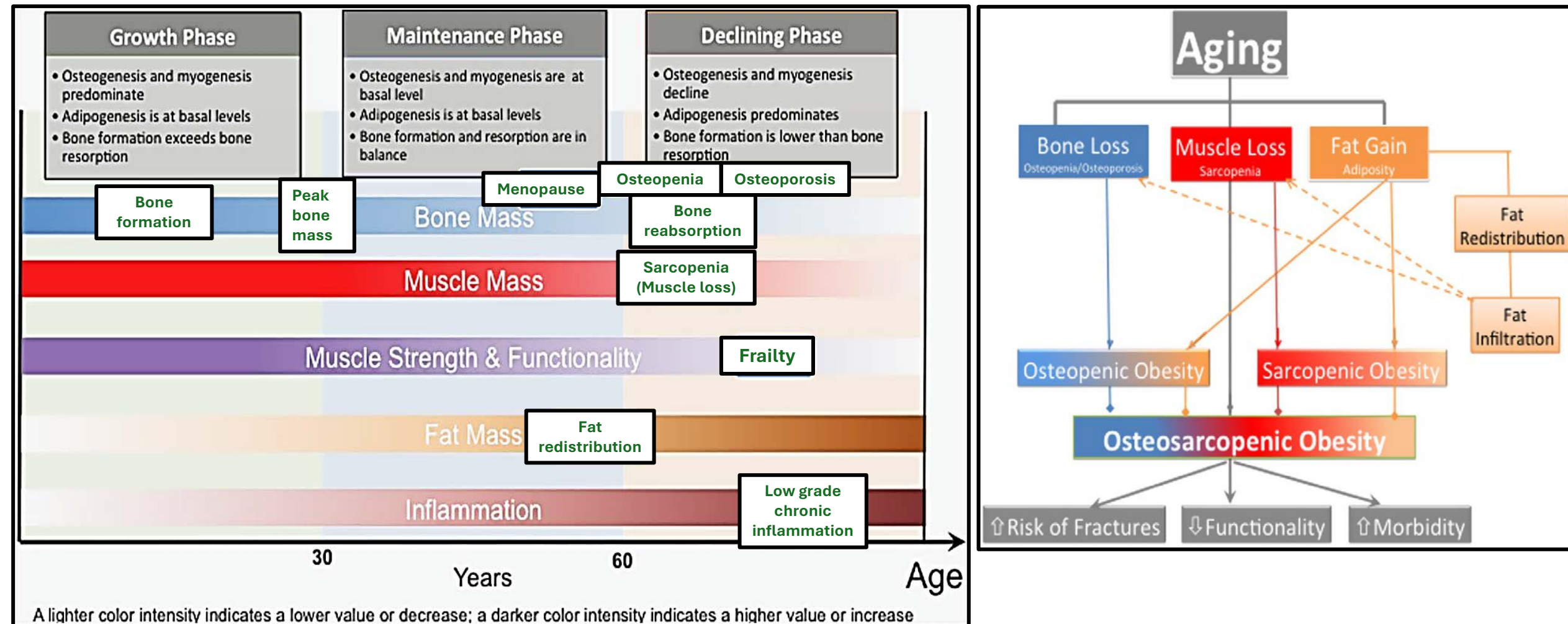
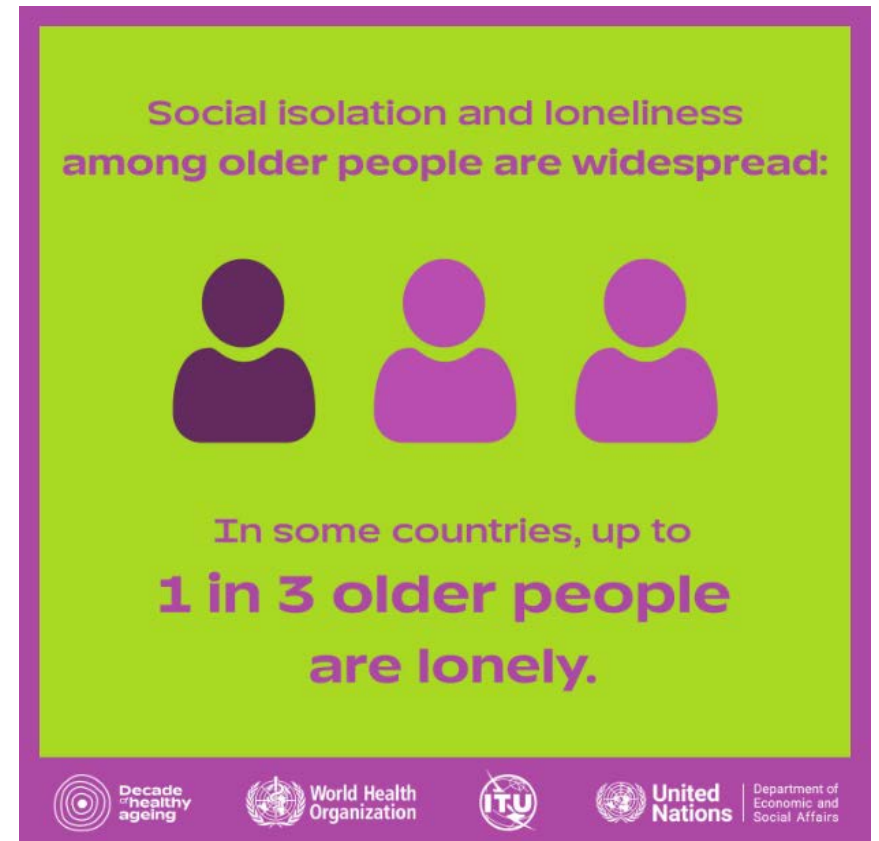


Figure 1 Aging human body: changes in bone, muscle and body fat with consequent changes in nutrient intake.
P. JafariNasabian et al. 2017 Journal of Endocrinology <https://doi.org/10.1530/JOE-16-0603>

Clinical case: Malnutrition in Elderly

Nutrition History (Brittany Bones):

- 1) Previously enjoyed cooking and eating but now finds meals tiring to prepare.
- 2) Often skips lunch (meals) or eats small portions
- 3) Drinks tea, crackers and microwave-ready foods
- 4) No alcohol, no smoking.
- 5) Lives and eats alone



Nutrient requirement in the elderly

- Increase in daily dietary protein, all dietary essential amino acids to prevent a negative nitrogen balance
- High in potassium, calcium, magnesium, zinc, selenium
- Vitamins B6 and B12, folic acid, vitamin C, vitamin D
- Omega-3 fatty acids (Unsaturated fatty acids)
- Dietary fiber (prevents constipation)
- Increase Water intake



Malnutrition in older adults

Diagnostic criteria:

- Unintentional weight loss
- Low body mass index or
- **Reduced muscle mass**

Etiology

- **Reduced food intake (Loss of taste, dysphagia, dementia, increased satiety)**
- **Decreased absorption**
- **Concurrent illnesses**
- **Immobility**

Risk factors for poor nutrition in elderly adults: **Social factors and Medical Factors**

Increased likelihood of isolation at mealtimes

Financial limitations affecting food acquisition

Dementia

Dentation

Dysgeusia-Impaired taste alter the cephalic phase of digestion

Dysphagia

Drugs

Physiologic risk factors for poor nutrition in elderly

- Decreased odor detection and lowers perceived odor intensity.
 - Decreased in the rate of gastric emptying
 - Increased in gastric pH
 - Decreased gastric parietal cell function (decreased HCl)
 - Increase in hormones and neurotransmitters thought to be involved in satiety.
 - Decreased stimulators involved in appetite



Clinical case (Brittany Bones): Malnutrition in Elderly

Investigations:

CBC: Mild anemia

Albumin: Low

Creatinine (Renal function): Normal

Vitamin B12: Borderline low

TSH: Normal

25-hydroxyvitamin D: Low →

DEXA scan: Osteopenia →

Handgrip test: Weak →

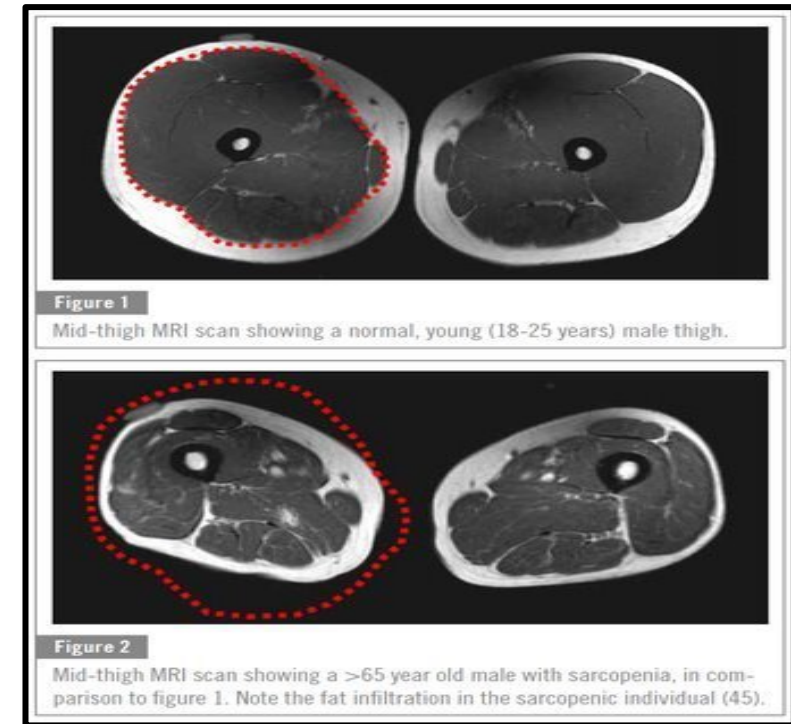
Gait speed: Slow →

BONE LOSS

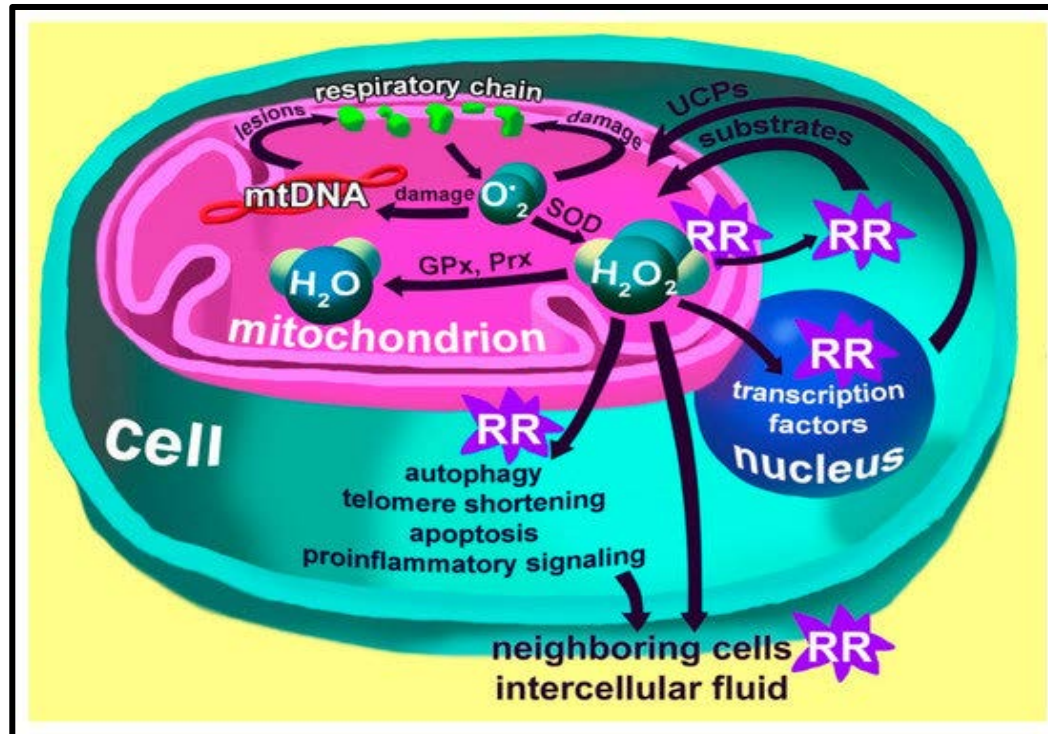
Sarcopenia

Sarcopenia → Frailty

- Decrease in appendicular muscle mass (2 SD below the mean for healthy adults)
- Measured by **dual-energy X-ray absorptiometry (DEXA)** or **bioelectrical impedance** in clinical practice
- Associated with:
 - Functional impairment
 - Disability and falls
 - Mortality
- Pathogenesis
 - Anabolic resistance of muscle
 - Mitochondrial reactive oxygen species (ROS)
 - Chronic low-grade inflammation
 - Intrinsic senescence → Muscle stem cell (satellite cell) dysfunction



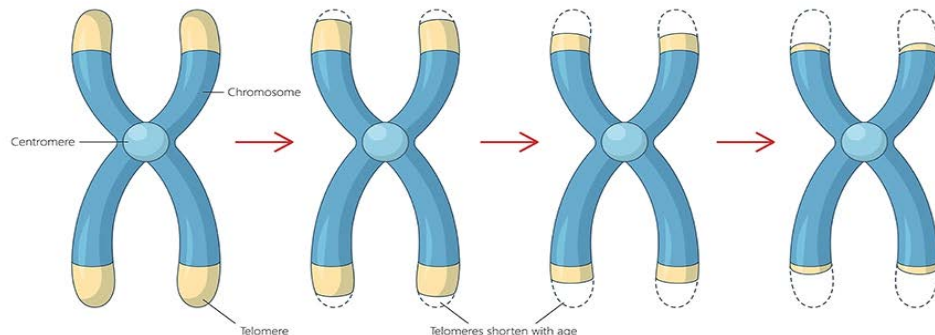
Oxidative stress and aging



Mitochondrial Aging causes excessive production of oxidants with decreased antioxidant production in elderly which can cause:

- 1) Autophagy
- 2) Potentially starts apoptosis
- 3) Increased expression of pro-inflammatory factors

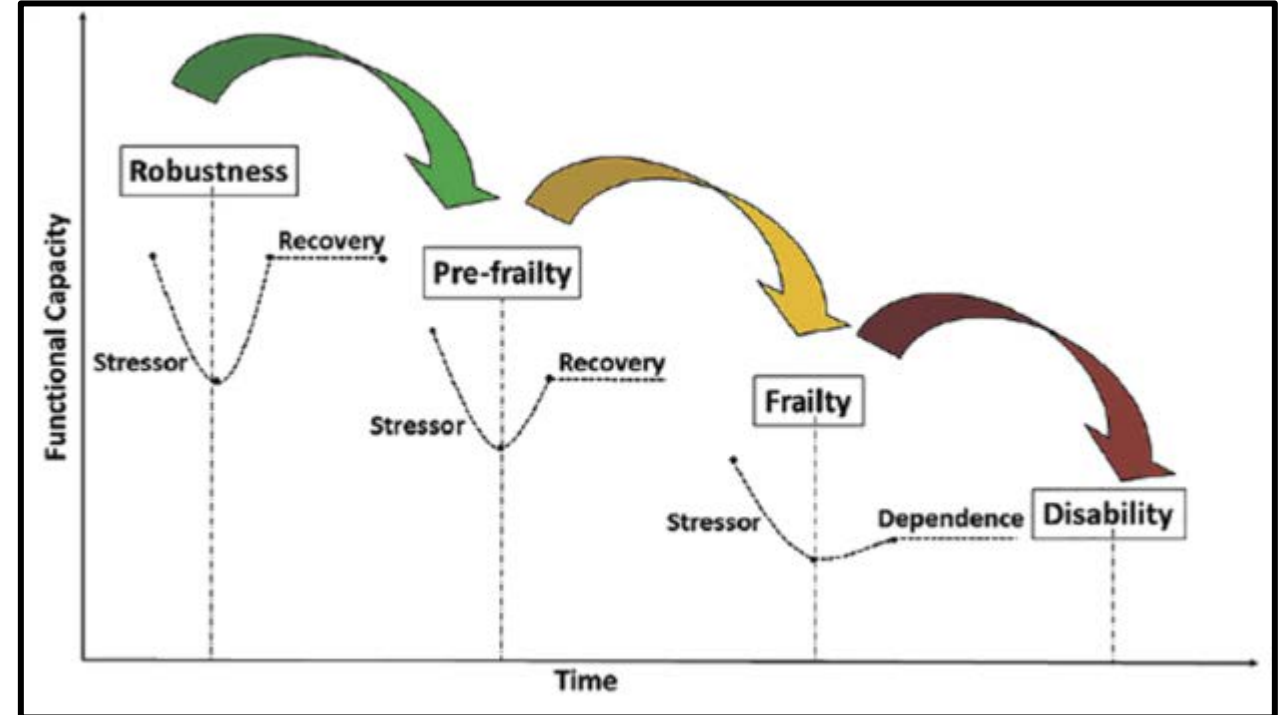
Telomere Shortening With Age



Shortening of the telomeres

Frailty

- An aging-related syndrome of **physiological decline**
- Frail older patients often present with an increased burden of symptoms, including
 - Weakness and fatigue
 - Falls and fractures
 - Reduced tolerance to medical and surgical interventions.



Anorexia vs Cachexia

Anorexia is a decrease in **appetite**, which naturally occurs in the elderly due to multiple physiologic changes. Such as:

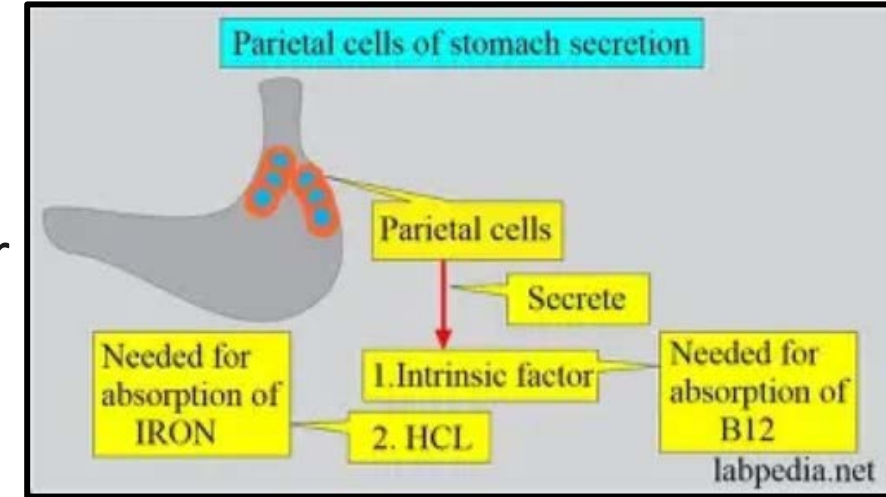
- 1) Decreased energy needs due to reduced physical activity
- 2) Decreased resting energy expenditure (REE)
- 3) Loss of lean body mass.

Cachexia is a complex metabolic syndrome associated with underlying illness and characterized by loss of muscle with or without loss of fat mass.

	Cachexia	Anorexia
Classification	Metabolic disorder associated with chronic inflammatory conditions	Eating disorder
Initially present with	Extensive loss of skeletal muscle mass; Possible loss of fat	Loss of adipose tissue; No loss of muscle mass
Nutritional supplementation	Does not reverse the condition	Reverses condition

MICRONUTRIENT DEFICIENCIES IN OLDER ADULTS

- **Vitamin B12 deficiency:** Higher prevalence of B12 deficiency in older adults (10 – 20%)
 - Loss of gastric mucosa, parietal cell, and intrinsic factor
- **Vitamin D deficiency** — Older individuals at higher risk for vitamin D deficiency include those who are institutionalized, homebound, have limited sun exposure, obesity, dark skin, osteoporosis, malabsorption or decreased kidney function.
- **Inadequate intake of calcium** — The efficiency of calcium absorption from the gastrointestinal tract decreases significantly after age 60 years old.
- **Multivitamin supplementation** — Many older adults use multivitamin supplements; however, only weak evidence for benefit for this practice. **A healthy, balanced diet of natural organic foods is advised.**



MANAGEMENT OF WEIGHT LOSS

- Treat underlying conditions
- Engage a nutritionist to reach calorie and protein requirements
- Physical exercise: Weight-bearing (resistance) training
- Endurance training



PROTEIN REQUIREMENT:

- At least **1 to 1.2 g/kg/day** protein to prevent lean muscle mass loss.
- During stress or injury, protein requirement may increase to **1.5 g/kg/day**.



Addressing inadequate food intake

- **Assistance with feeding or shopping**

Ensure that feeding or shopping assistance is available. Social work support if inadequate finances.

- **Food preferences**

Suggest foods that fit the patient's ethnic or regional preferences.



- **Dietary supplementation**

Consider ways to supplement the patient's diet. Increase the nutrient density of food.

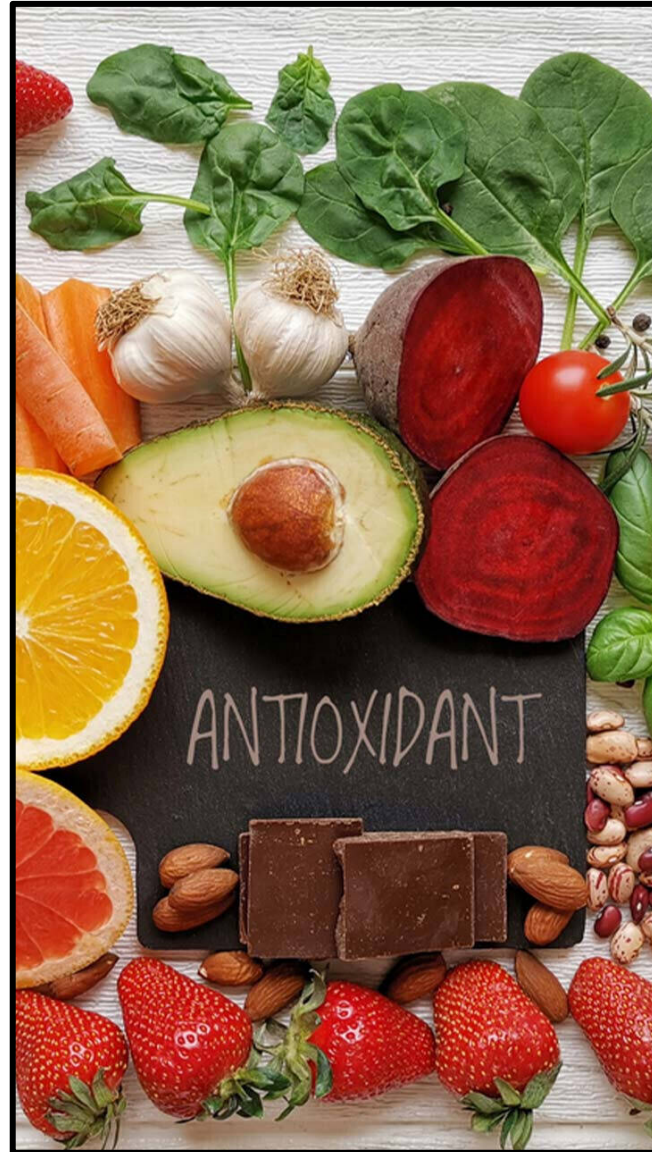
Give a daily multivitamin and mineral supplement until the cause of inadequate intake is determined.

Consider a liquid dietary supplement.



Antioxidant rich foods for elderly

- Berries
- Leafy Greens
- Orange and Red vegetables
- Nuts and Seeds
- Dark Chocolate (>70% cocoa)
- Citrus Fruits
- Grapes (especially red and purple)
- Tomatoes
- Beans and Legumes
- Green Tea



Summary

- Elderly are a growing but vulnerable group in our population.
- Social, medical, and physiological causes that predispose elderly patients to malnutrition.
- Malnutrition contributes to morbidity and mortality
- Explain the terms sarcopenia, frailty, anorexia, and cachexia
- Highlight endocrine, physiological, and biochemical changes in aging
- Management and prevention of chronic disease requires a holistic approach, including diet and lifelong physical activity
- Optimize diet with nutrient-dense foods, then consider supplementation when appropriate.